Divided Differences

Let $x_0, x_1, \ldots, x_n$ be distinct points, we define the 0th divided difference at x_i by $f[x_i] = f(x_i)$, the first divided difference at $[x_i, x_j]$ is,

$$f[x_i, x_j] = \frac{f(x_i) - f(x_j)}{x_i - x_j} = \frac{f[x_i] - f[x_j]}{x_i - x_j}$$ in general, the kth divided difference

$$f[x_0, x_1, \ldots, x_k] = \frac{f[x_0, x_1, \ldots, x_{k-1}] - f[x_1, \ldots, x_k]}{x_0 - x_k}$$

Ex (1): Find the d. d. $f[x_0, x_1], f[x_0, x_1, x_2], f[x_0, x_2, x_1], f[x_1, x_0]$

i	0	1	2
x_i	1	2	3
$f(x_i)$	3	5	7

Solu.

$$f[x_0, x_1] = \frac{f(x_0) - f(x_1)}{x_0 - x_1} = \frac{3-5}{1-2} = 2$$

$$f[x_1, x_0] = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{5-3}{2-1} = 2$$

$$f[x_0, x_1, x_2] = \frac{f[x_0, x_1] - f[x_1, x_2]}{x_0 - x_2} = \frac{2-2}{1-3} = 0$$

$$f[x_1, x_2] = \frac{f(x_1) - f(x_2)}{x_1 - x_2} = \frac{5-7}{2-3} = 2$$

$$f[x_0, x_2, x_1] = \frac{f[x_0, x_2] - f[x_2, x_1]}{x_0 - x_1} = 0$$

First divided difference

$$f[x_i, x] = \frac{f(x_i) - f(x)}{x_i - x}$$

$$f(x) = f(x_i) + (x - x_i) f[x_i, x]$$

$$f(x) = f(x_0) + (x - x_0) f[x_0, x]$$

$$p_1(x) = f(x_0) + (x - x_0) f[x_0, x]$$

Approximate $f(x)$ and interpolation $f(x)$ at (x_0, x_1), $p_1(x_0) = f(x_0)$

$$p_1(x_1) = f(x_0) + (x_1 - x_0)\frac{f(x_0) - f(x_1)}{x_0 - x_1} = f(x_0) + (x_1 - x_0)\frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$= f(x_0) + f(x_1) - f(x_0) = f(x_1)$$

Ex (2): Let $(x_i, f(x_i))$ be given in the following table.

i	x_i	$f(x_i)$	1st d. d.	2nd d. d.	3rd d. d.
0	1	2			
			3		
1	2	5		1	
			5		0
2	3	10		1	
			7		
3	4	17			

Estimate $f(1.5)$ using Newton d.d. interpolation poly.

$$f[x_0, x_1] = \frac{f[x_0] - f[x_1]}{x_0 - x_1} = \frac{f(x_0) - f(x_1)}{x_0 - x_1} = \frac{2-5}{1-2} = 3$$

$$f[x_1, x_2] = \frac{f[x_1] - f[x_2]}{x_1 - x_2} = \frac{5-10}{-1} = 5$$

$$f[x_2, x_3] = \frac{f[x_2] - f[x_3]}{x_2 - x_3} = \frac{10-17}{-1} = 7$$

$$f[x_0, x_1, x_2] = \frac{f[x_0, x_1] - f[x_1, x_2]}{x_0 - x_2} = \frac{3-5}{1-3} = 1$$

$$f[x_1, x_2, x_3] = \frac{f[x_1, x_2] - f[x_2, x_3]}{x_1 - x_3} = \frac{5-7}{2-4} = 1$$

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_0, x_1, x_2] - f[x_1, x_2, x_3]}{x_0 - x_3} = \frac{1-1}{1-4} = 0$$

$$p(x) = f(x_0) + (x - x_0) f[x_0, x_1] + (x - x_0)(x - x_1) f[x_0, x_1, x_2]$$

$p_2(x) \cong f(x)$ then to estimate $f(x = 1.5)$, we evaluate $p(x = 1.5)$

$$\Rightarrow p(x = 1.5) = f(x_0) + (1.5 - x_0) f[x_0, x_1] + (1.5 - x_0)(1.5 - x_1) f[x_0, x_1, x_2]$$

$$= 2 + (0.5)3 + (0.5)(-0.5)1 = 2 + \frac{3}{2} - \frac{1}{4} = \frac{8}{4} + \frac{6}{4} - \frac{1}{4} = \frac{13}{4} = 3.25$$

Newton Forward divided difference

$x_0, x_1, \ldots, x_n$

$$f(x) \cong p_n(x) = f[x_0] + (x - x_0) f[x_0, x_1] + \ldots + f[x_0, \ldots, x_n](x - x_0) \cdots (x - x_{n-1})$$

$x_0, x_1, \ldots, x_n$ are arranged consecutively with equal spacing.

Let $h = x_{i+1} - x_i \Rightarrow x_1 - x_0 = h$

$$x_2 - x_1 = h$$

$$x_3 - x_2 = h$$

$x_0 = 0, h = 1$

$x_1 = 1, x_2 = 2, x_3 = 3$ equal spacing points

$x = x_0 + sh$

$x - x_0 = sh + x_0 - x_0 = sh \Rightarrow (x - x_1) = (s - 1)h$

$(x - x_0)(x - x_1) = s(s - 1)h^2$

Let $x - x_i = (s - i)h, i = 0,1,\ldots,n$

$x - x_0 = (s - 0)h = sh$

$x - x_1 = (s - 1)h$

$x - x_2 = (s - 2)h$

$x - x_3 = (s - 3)h$

$$\Rightarrow p_n(x) = p_n(x_0 + sh) = f[x_0] + shf[x_0, x_1] + s(s - 1)h^2 f[x_0, x_1, x_2] + \cdots + s(s - 1)$$

$$\cdots (s - n + 1)h^n f[x_0, x_1, \ldots, x_n] = \sum_{k=0}^{n} s(s - 1) \cdots (s - k + 1)h^k f[x_0, x_1, \ldots, x_k]$$

$$\because \binom{s}{k} = \frac{s(s-1)\cdots(s-k+1)}{k!}$$

$$p_n(x) = \sum_{k=0}^{n} \binom{s}{k} k!\, h^k f[x_0, x_1, \ldots, x_k] \text{ (N. F. d. d. formula)}$$

Newton Backward divided difference

$x_0, x_1, \ldots, x_n$

The interpolating point $x_0, x_1, \ldots, x_n$ are reordered as $x_n, x_{n-}, \ldots, x_0$ a formula.

$$p_n(x) = f[x_n] + f[x_{n-1}, x_n](x - x_n) + f[x_{n-2}, x_{n-1}, x_n](x - x_n)(x - x_{n-1}) + \ldots +$$

$$f[x_0, x_1, \ldots, x_n](x - x_n)(x - x_{n-1})\cdots(x - x_1)$$

$x = x_n + sh$ and $x = x_i + (s + n - i)h$

$$p_n(x) = p_n(x_n + sh) = f[x_n] + shf[x_{n-1}, x_n] + s(s+1)h^2 f[x_{n-2}, x_{n-1}, x_n] + \ldots + s(s+1)$$

$$(s+2)\cdots(s+n-1)h^n f[x_0, x_1, \ldots, x_n] \text{(N. B. d. d. formula)}$$

Ex (3): estimate $f(2.5)$

i	0	1	2	3
x_i	0	1	2	3
$f(x_i)$	2	6	12	20

i	x_i	$f(x_i)$	1st d. d.	2nd d. d.	3rd d. d.
0	0	2			
			4		
1	1	6		1	
			6		0
2	2	12		1	
			8		
3	3	20			

N. f. $f(2.5) = 2 + 4(x-0) + 1(x-0)(x-1) = 2 + 4(2.5) + (2.5)(2.5-1) = 15.75$

N. b. $f(2.5) = 20 + 8(2.5-3) + 1(2.5-3)(2.5-2) = 15.75$

Ex (4):

i	x_i	$f(x_i)$	1st d. d.	2nd d. d.	3rd d. d.	4th d. d.
0	1.0	0.76519				
			-0.4837			
1	1.3	0.62008		-0.108733		
			-0.54894		0.06587	
2	1.6	0.4554		-0.049443		0.0018
			-0.57861		0.06806	
3	1.9	0.281		0.01181		
			-0.57152			
4	2.2	0.11036				

1. If an approximation of $f(1.1)$ is required using N. F. d. d. formula.

$$x_0, x_1, x_2, x_3, x_4, h = x_{i+1} - x_i = 0.3, x = x_0 + sh \Rightarrow x = 1.5 + s(0.3)$$

$$\Rightarrow 1.1 = 1.0 + s(0.3) \Rightarrow \frac{(1.1-1.0)}{0.3} = s = \frac{1}{3}$$

$$p_4(x) = p_4(x_0 + sh) = p_4\left(1.0 + \frac{1}{3}0.3\right) = f[x_0] + shf[x_0, x_1] + s(s-1)h^2 f[x_0, x_1, x_2]$$

$$+ s(s-1)(s-2)h^3 f[x_0, x_1, x_2, x_3] + s(s-1)(s-2)(s-3)h^4 f[x_0, x_1, x_2, x_3, x_4]$$

$$= 0.765 + \left(\frac{1}{3}\right)(0.3)(-0.4837) + \frac{1}{3}\left(-\frac{2}{3}\right)(0.3)^2(-0.10873) + \frac{1}{3}\left(-\frac{2}{3}\right)\left(\frac{-5}{3}\right)(0.3)^3$$

$$(0.06587) + \frac{1}{3}\left(-\frac{2}{3}\right)\left(\frac{-5}{3}\right)\left(\frac{-8}{3}\right)(0.3)^4(0.0018) = 0.719698$$

2. If an approximation of $f(2)$ is required using N. B. d. d. formula reared the points $x_n, x_{n-1}, \ldots, x_0 \Rightarrow [x_4, x_3, x_2, x_1, x_0]$.

$$x = x_4 + sh \Rightarrow 2.0 = 2.2 + s(0.3) \Rightarrow s = \frac{-0.2}{0.3} = \frac{-2}{3} = -0.6666$$

$$p_4(x) = p_4(x_4 + sh) = p_4\left(2.2 + \left(\frac{-2}{3}\right)0.3\right) = f[x_4] + shf[x_4, x_3] + s(s+1)h^2$$

$$f[x_4, x_3, x_2] + s(s+1)(s+2)h^3 f[x_4, x_3, x_2, x_1] + s(s+1)(s+2)(s+3)h^4$$

$$f[x_4, x_3, x_2, x_1, x_0] = 0.11036 + \left(\frac{-2}{3}\right)(0.3)(-0.57152) + \left(\frac{-2}{3}\right)\left(\frac{1}{3}\right)(0.3)^2(0.01181)$$

$$+ \left(\frac{-2}{3}\right)\left(\frac{1}{3}\right)\left(\frac{4}{3}\right)(0.3)^3(0.06806) + \left(\frac{-2}{3}\right)\left(\frac{1}{3}\right)\left(\frac{4}{3}\right)\left(\frac{5}{3}\right)(0.3)^4(0.0018) = 0.2238$$